



Ideation Proposal

GUJCOST-ROBOFEST 5.0

IDEATION PROPOSAL

Aerial Robotics: Minefield Navigation Challenge Senior Category



AJAY KUMAR GARG ENGINEERING COLLEGE

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1.0 Executive Summary

This document outlines the operational strategy for the **Minefield Navigation Challenge** at Robofest 5.0. Our approach leverages a **four-drone autonomous swarm** operating under a **Specialized Parallel Survey Strategy**. Each drone is assigned a unique role—Surveyor, Scout, and Leader—enabling simultaneous detection of static obstacles (poles) and dynamic hazards (simulated mines).

The mission execution is structured across **five core phases**:

1. **Pre-Mission Field Calibration** – Establishes a GPS-based World Coordinate System (WCS).
2. **Autonomous Swarm Initialisation** – Real-time localisation and swarm rendezvous.
3. **Coordinated Task Execution** – Parallel mapping of poles and mines.
4. **Real-Time Data Fusion & Path Planning** – Creation of Master Map and optimal pathfinding.
5. **Human Guidance & Mission Completion** – Guiding the person-at-risk safely to the exit zone.

This strategy ensures **time efficiency, precision, and robustness**, improving our chances of a higher **Time Bonus and Mission Accuracy Score**. Weight and power budgets were analysed to confirm endurance above 15 minutes in real-world conditions, ensuring feasibility within competition constraints (see Annexures I).

2.0 Type of Robot

- **System Type:** Multi-agent Drone Swarm (4 UAVs)
- **Mission Goal:** Detect and localise poles & mines, generate optimal paths, and guide a human through the minefield.
- **Operational Area:** 100m × 20m competition field

2.1 Size of Robot

Size of robot proposed for Proof of concept (Small version):

a. Length: 223.54mm b. Width: 223mm. c. Height: 39mm

Size of robot proposed as Prototype (Actual Version)

a. Length: 220mm b. Width: 220mm c. Height: 39mm

3.0 Robot Assembly Design

12. Each drone is designed for a **specialised role**:

1. **Surveyor Drone** – High-altitude fast mapping of poles.
2. **Scout Drone A** – Low-altitude mine detection (Zone 1).
3. **Scout Drone B** – Low-altitude mine detection (Zone 2).
4. **Leader Drone** – Swarm coordinator, data fusion, path planning, and human interaction.

4.0 Components to be Used

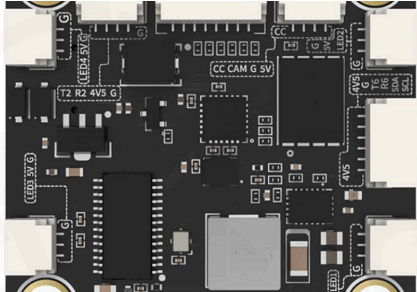
4.1 Structural Components

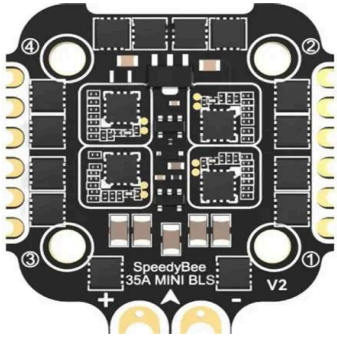
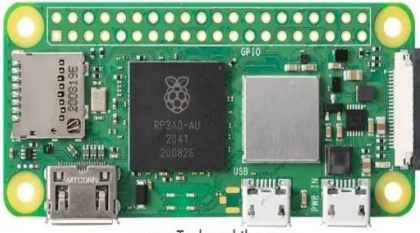
Component	Specification	Quantity
Custom Carbon Fibre Frame	300*300mm	4
Nylon Spacers	50mm	48
M2, M3 size bolts	Holding ESC, FLIGHT CONTROLLER	48
3D Printed parts	Battery mount, Raspberry Pi plate	4

4.2 Propulsion System

Component	Specification	Quantity
Brushless DC Motors	iFlight XING2 1404	16
Propellers		16

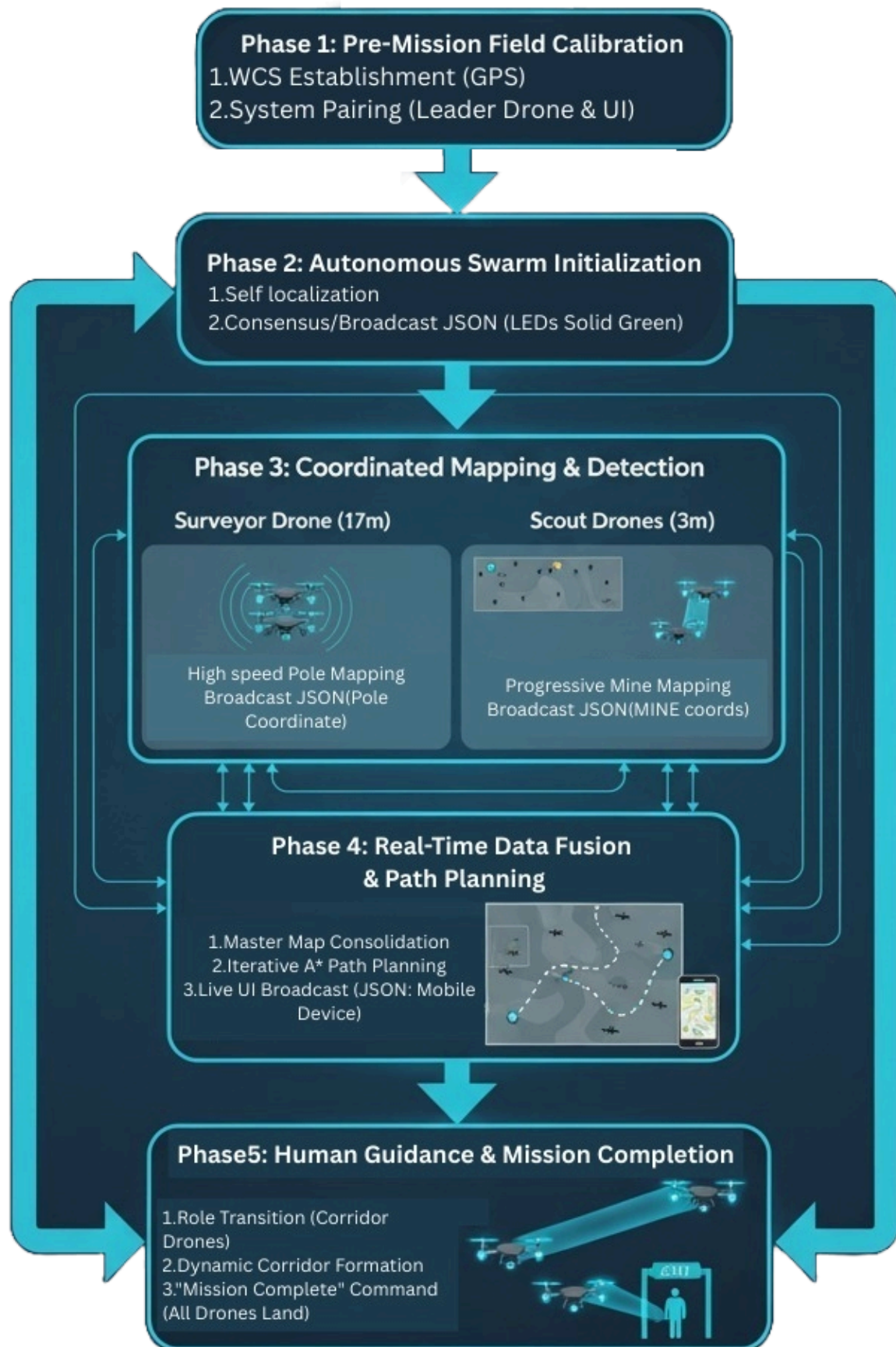
4.3 Electronic Components

Component	Specification	Quantity
SpeedyBee 	STM32F405, a high-speed ARM microcontroller for efficient processing	4
RPi Camera Module 3 	12MP IMX708 sensor, 4608×3456 resolution, 76° FOV	4
Molicel P28A 18650 pack, 2800 mAh 	2800mAh, 3.7V nominal, high discharge 35A, Li-ion battery	4

35A 2-5S BLHELI S 4 	35A Continuous current, 2-5S LiPo compatible, BLHeli_S firmware	4
Benewake TFmini-S LiDAR 	12m range, $\pm 1\%$ accuracy, I2C/UART interface	4
GPS (M100 Pro) 	High-precision GNSS supports GPS/GLONASS/Beidou	4
Raspberry Pi Zero 2 W 	Quad-core 64-bit ARM Cortex-A53 512 MB LPDDR2, Wi-Fi 802.11 b/g/n + BT 4.2 (BLE), CSI-2 camera,	4

5.0 Mission Execution Methodology

The mission is executed in five distinct and sequential phases. This methodology integrates perception, planning, and communication to ensure a rapid, precise, and robust operation.



5.1 Pre-Mission Calibration & WCS Definition

This one-time, offline operator task establishes the "ground truth" for the entire mission.

- **WCS Establishment:** The **World Coordinate System (WCS)** is established by **pre-programming the absolute GPS coordinates of the four corners of the competition field** into all drones.
- **System Pairing:** The Leader drone's local Wi-Fi network is paired with the person-at-risk's mobile device to enable a one-way data stream for the live UI.

5.2 Autonomous Swarm Initialisation & Consensus

This phase is performed autonomously on the ground to ensure a valid and safe starting configuration.

- **Self-Localisation:** Each drone determines its precise starting position and heading within the WCS using its GPS and IMU.
- **Swarm Rendezvous & Data Broadcast:** A swarm-wide communication link is established over Wi-Fi. **Each drone broadcasts a JSON packet containing its unique ID and initial pose estimate.**
- **Consensus Validation:** The Leader drone validates swarm readiness by performing a consensus check on the received data. It then issues the synchronised "take-off clear" signal.

5.3 Coordinated Mapping & Detection

This phase employs a parallel and progressive strategy to build the map as quickly as possible.

- **Surveyor Mission:** The Surveyor drone ascends to **17 meters** and performs its high-speed survey. After processing each image, **it broadcasts a JSON packet containing a list of all newly discovered pole GPS coordinates** to the swarm.
- **Scout Mission (Progressive Mapping):** The two Scout drones ascend to **3 meters** and map the front half of the minefield. As they detect mines, **they continuously broadcast JSON packets with the georeferenced GPS coordinates of the mines** to the swarm.
- **Onboard Processing:** All target detection and georeferencing is performed onboard each drone before the coordinates are broadcast.

5.4 Real-Time Data Fusion & Path Planning

This phase runs concurrently with mapping to minimize delay.

- **Master Map Consolidation:** The Leader drone listens for all incoming data packets from the Surveyor and Scouts and uses the coordinates to continuously update its unified Master Map.
- **Iterative Path Computation:** The A* algorithm runs iteratively on the live Master Map to find the optimal safe path.

- **Live UI Broadcast:** Throughout this phase, the **Leader drone continuously broadcasts the state of the entire system**—including the updated Master Map, the positions of all drones, and the current best-path solution—in a **JSON packet to the user's mobile device** every 1-2 seconds.



Fig 1.3: Proposed Live Monitoring Dashboard
Please Follow this link for Web Simulated Prototype(eagle-ayush.netlify.app)

The proposed system includes a live monitoring dashboard designed for intuitive, real-time mission oversight. The left panel features a **Live Map**(<https://eagle-ayush.netlify.app>) of the 90m×20m competition field, dynamically updating to show the location of detected poles and mines. This map is generated by the leader drone, which consolidates data from all scout and surveyor drones. This dashboard serves as a central point for data fusion and human guidance, ensuring that the person-at-risk can be guided safely along the optimal path.

5.5 Human Guidance & Mission Completion

This final phase begins as soon as a safe path is locked in.

- **Role Transition & Path Dissemination:** The Leader **broadcasts the final path waypoints** to the Scout drones, which transition to their "Corridor Drone" role.
- **Dynamic Corridor Formation:** The Leader drone acts as a "pacer" while the two Corridor Drones form a **~2-meter-wide "corridor of light"** using their LEDs.
- **Completion:** Upon successful exit, the **Leader drone broadcasts a final "Mission Complete" command**, instructing all drones to autonomously return to the Start Zone and land.

6.0 Algorithmic Framework

6.1 Core Algorithms of the Autonomous Swarm

The swarm's intelligent behaviour is governed by a suite of core algorithms, orchestrated by the ROS (Robot Operating System) framework on the onboard Raspberry Pi Zero 2 W. This section details the input, process, and output of each major algorithmic component, along with the strategic justification for its design.

6.2 Swarm Initialisation Algorithm (Consensus-Based)

- **Input:** Real-time GPS and IMU data from all four drones; pre-programmed GPS coordinates of the field corners.
- **Process:**
 1. Each drone broadcasts its initial pose estimate in a JSON packet.
 2. The Leader drone performs a **consensus-based cross-validation, verifying that all drones are online and that their reported positions are unique and fall within the predefined Start Zone boundaries.**
- **Output:** A confirmed, shared World Coordinate System (WCS) and a synchronised "take-off clear" signal. This is physically indicated by all drones switching their LEDs to a **solid green** state.
- **Justification:** This consensus mechanism serves as a critical failsafe, preventing the mission from starting with a faulty GPS or communication link, thereby ensuring a robust and reliable initial state for the entire swarm.

6.3 Perception Algorithm (OpenCV Vision Pipeline)

- **Input:** A high-resolution image frame from the Pi Camera 3.
- **Process (Batch-Optimized):**
 1. **Correction:** Apply lens distortion correction using pre-calibrated camera intrinsic parameters.
 2. **Colour Segmentation:** Convert the frame to the HSV colour space to enable robust colour-based filtering.

3. **Masking:** Apply optimised colour masks to isolate all potential pole and mine targets simultaneously.
 4. **Contour Extraction:** Use `cv2.findContours()` to extract the outlines of all detected colour blobs.
 5. **Filtering:** Filter contours based on area, circularity, and other shape metrics to discard noise and validate true targets.
- **Output:** A list of valid pixel coordinates (u, v) for *all* detected targets within the single image.
 - **Justification:** The **batch-optimised** approach processes all targets in one pass, minimising redundant calculations and CPU cycles, which saves critical mission time at each survey waypoint.

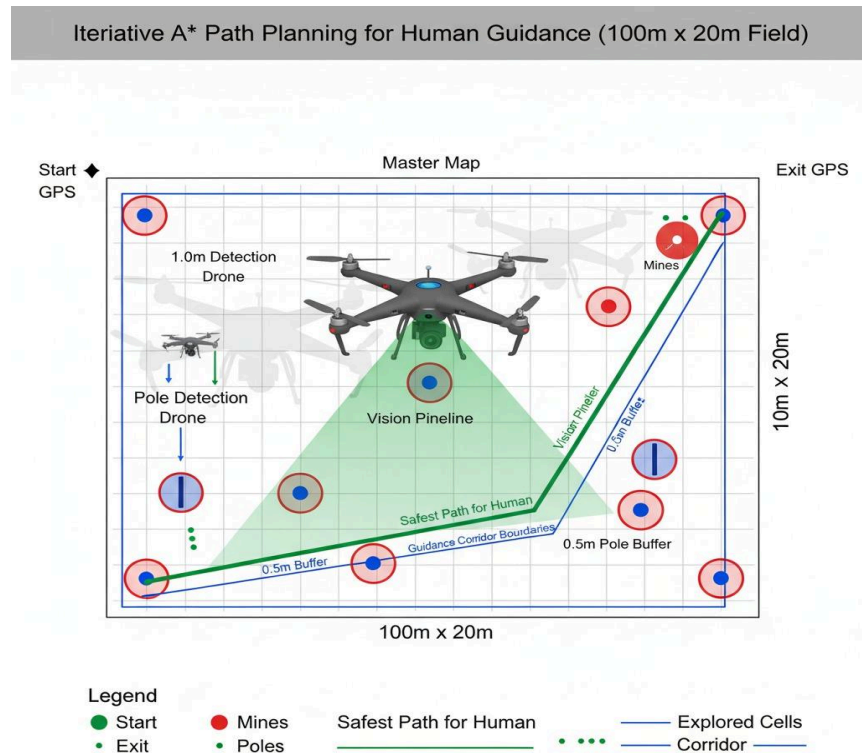
6.4 Georeferencing Algorithm (Photogrammetric Triangulation)

- **Input:** A list of pixel coordinates (u, v) , the drone's precise GPS position, altitude (from LiDAR), attitude (from IMU), and camera intrinsics.
- **Process:**
 1. A 3D direction vector is generated from the camera's centre through each target pixel.
 2. This vector is rotated into the global World Coordinate System (WCS) using the drone's attitude (yaw, pitch, roll) matrix.
 3. The final GPS coordinate is calculated by finding the intersection of this rotated vector with the ground plane.
- **Output:** A list of absolute GPS coordinates (Lat, Lon) for each target.
- **Justification:** This algorithm leverages a **fusion of multiple sensors (GPS, LiDAR, IMU)**, offering a high degree of accuracy that is resilient to drift from a single sensor. The use of LiDAR for altitude is especially critical for precise scaling.

6.5 Path Planning Algorithm (Iterative A*)

- **Input:** A real-time Master Map with obstacles (poles with a 0.5m buffer, mines with a 1.0m buffer), and the Start/Exit GPS coordinates.
- **Process:**
 1. The Master Map is converted into a 2D occupancy grid.
 2. The A* algorithm searches for the lowest-cost path, avoiding all "no-go" obstacle cells.
 3. **Iterative Execution:** The algorithm runs repeatedly as new map data arrives.
- **Output:** An ordered list of GPS waypoints that define the shortest, safest route.

- **Justification:** The **iterative approach** is the core of our **Progressive Mapping** strategy. It allows the system to find a viable path based on a partial map, enabling the guidance phase to begin the moment a path is found and significantly reducing the user's wait time.



6.6 Guidance Algorithm (Synchronised Formation Control)

- **Input:** The final list of path waypoints from the Leader; real-time human gestures (*forward, pause*) detected by the Leader's camera.
- **Process:**
 1. The Scout drones cease mapping and transition to their "**Corridor Drone**" role.
 2. The Leader drone acts as a "pacer," following the waypoints at a steady pace that is **dynamically controlled by the user's gestures**.
 3. Each Corridor Drone calculates its target position by taking the Leader's current waypoint, determining a vector perpendicular to the path's direction, and offsetting itself to form a ~2-meter-wide lane.
- **Output:** A dynamic, moving "**corridor of light**" that provides unambiguous, interactive, and physically bounded guidance to the person-at-risk.

7.0 Calculation of Optimal Flight Altitudes

7.1. Surveyor Drone Altitude (Optimised for Efficiency)

- **Objective:** Rapidly map large obstacles, **Assumption**(10cm pole bases) with high reliability.

- **Precision Goal:** A 10cm target must appear as a feature of at least **15x15 pixels**.
- **Calculation:**
 - Required Ground Sample Distance (GSD): **0.1m / 15px = 0.0067 m/px**
 - Maximum Altitude (**H_{max}**) for this GSD:

$$H_{max} = (0.0067 * 4608) / (2 * \tan(76^\circ/2)) \approx 19.8 \text{ meters}$$
 - Operational Altitude (with a 15% safety margin): $19.8\text{m} * 0.85 \approx 16.8 \text{ meters}$
- **Final Altitude: 17 meters.** This provides fast area coverage while ensuring pole detection is highly reliable.

7.2. Scout Drone Altitude (Optimised for Precision)

- **Objective:** Map critical targets (40cm mines) with the highest possible accuracy for safe path planning.
- **Precision Goal:** To achieve sub-centimetre accuracy for mine centre-points.
- **Calculation** (A low, stable altitude is chosen to maximise precision):
 - **Selected Operational Altitude: 3 meters**
 - GSD at this altitude:

$$\text{GSD} = (2 * 3\text{m} * \tan(76^\circ/2)) / 4608 \approx 0.001 \text{ m/px (or } \mathbf{0.1 \text{ cm/pixel}})$$
 - Resulting precision: A 40cm mine will be rendered as a **~400-pixel diameter circle**, far exceeding requirements and guaranteeing exceptional accuracy.
- **Final Altitude: 3 meters.** This prioritises data quality for the most critical mapping task.

8.0 Timeline

	Task	Start Date	End Date
1	Research	11/07/2025	20/07/2025
2	Analysis of the available Minefield Navigation Challenge	21/07/2025	31/07/2025
3	Rough Sketch	01/08/2025	05/08/2025
4	Hardware and Software Architecture	06/08/2025	10/08/2025
5	Selection of Electronic and Mechanical Components	06/08/2025	10/08/2025
6	Designing of Components	11/08/2025	15/08/2025
7	Assembling of Components	15/08/2025	20/08/2025
8	Error Refining and Redesigning	15/08/2025	06/08/2025
9	Model Finalization	15/08/2025	06/08/2025

10	Documentation	12/07/2025	06/08/2025
11	Submission at GujCost for Level I	07/09/2025	08/09/2025
12	Simulation of Design	10/09/2025	20/09/2025
13	Ordering of Material for Prototype	26/09/2025	30/09/2025
14	Assembly of Electronics Components on Prototype	01/10/2025	10/10/2025
15	Software work	11/10/2025	10/11/2025
16	Testing of Prototype	01/10/2025	14/11/2025
17	Required Modification	01/10/2025	14/11/2025
18	Testing and Troubleshooting	01/10/2025	14/11/2025
18	Final Approval by Mentors	01/11/2025	14/11/2025
20	Submission at GujCost for level II	14/11/2025	15/11/2025
21	Preparation for 3rd Stage	16/11/2025	10/12/2025
22	Working on Final Prototype	16/11/2025	09/12/2025
23	Required Modification	16/11/2025	09/12/2025
24	Final Approval by Mentors	01/12/2025	08/12/2025
25	Submission at GujCost level III	09/12/2025	10/12/2025
26	Robot Demo Optimisation	09/12/2025	10/12/2025

9.0 Justifications and Optimization

Justifications for Electronic Components

S.NO	Components	Justifications
1.	SpeedyBee	Their advanced processors and all-in-one designs provide smooth flight control and easy wireless configuration. Built-in safety features and robust ESCs ensure reliable performance for small, high-agility builds
2.	Raspberry Pi Zero 2 W	Chosen as the onboard companion computer to perform all high-level mission logic and advanced computer vision tasks (pole

		& mine detection) using the OpenCV library. Its strong community support ensures rapid development
3.	Camera Module 3	Provides wide-FOV imagery for accurate pole and mine detection at both high and low altitudes.
4.	Battery (4S Li-ion P28A 18650)	Balances endurance (~15+ min) with weight compliance (<500 g AWW).
5.	Sensors (LiDAR, GPS, IMU)	LiDAR ensures accurate obstacle ranging, GPS provides global position, and IMU supports real-time orientation tracking.
6.	BEC (5V/5A)	Provides a clean, regulated power supply for Raspberry Pi and sensors, preventing brownouts.

Justifications for Other Accessories

S.No.	Components	Justifications
1.	Wiring Harnesses & Connectors	Lightweight and optimized to reduce unnecessary weight while ensuring secure connections.

Optimizations Implemented

- **Batch Geotagging:** Reduces redundant calculations at each waypoint, saving 5–10s total mission time.
- **Efficient Communication Packets:** Broadcasts lists instead of individual coordinates, reducing RF congestion.

10.0 Application in Societal Context

This system has significant applications in humanitarian and defense contexts, such as minefield clearance, disaster response, and autonomous navigation in hazardous environments where human access is unsafe.

- **Humanitarian Demining:** Drones can safely map landmines and unexploded bombs, guiding disposal teams and protecting lives. This makes previously unsafe areas habitable again.
- **Disaster Response:** A drone swarm can quickly survey disaster zones, identifying safe routes and locating survivors during events like earthquakes, floods, or wildfires. This speeds up rescue efforts and saves lives.

- **Precision Agriculture:** The drones can spot crop problems early using their cameras, which helps farmers save water and chemicals by treating only what's needed, thus improving crop yield.
- **Infrastructure Inspection:** Autonomous drones can check bridges, pipelines, and power lines for damage. This keeps human workers safe and prevents costly infrastructure failures.



Humanitarian Demining

Drones map landmines and unexploded bombs, guiding disposal teams and protecting lives, making areas safe again.



Disaster Response

Swarms quickly survey disaster zones, finding safe routes and locating survivors to speed up rescue efforts during earthquakes, floods, or wildfires.



Precision Agriculture

Precision Agriculture: Drones spot crop problems early, helping farmers save water and chemicals by treating only what's needed, improving crop yield.



Infrastructure Inspection

Autonomous drones check bridges, pipelines, and power lines for damage, keeping human workers safe and preventing costly infrastructure failures.

11.0 Future Scope of Development

To enhance the robot's capabilities beyond the competition, future development can focus on three key areas:

- **Advanced Sensing and Classification:** Upgrading the drones with multispectral or hyperspectral cameras would allow them to identify objects based on chemical signatures. This would enable them to distinguish real landmines from other metallic objects, a critical feature for humanitarian demining. This technology could also be applied to agriculture for identifying stressed crops with greater precision.
- **Enhanced Autonomy and Swarm Intelligence:** Instead of having fixed roles, the swarm could implement dynamic role assignment. The drones could automatically re-assign tasks if one malfunctions or runs low on power, making the system more robust and reliable. Machine learning models could be integrated to classify different types of mines or unexploded ordnance with greater accuracy.
- **Expansion to Broader Applications:** The core technology of a multi-drone swarm for mapping and navigation can be adapted for new societal applications. This includes autonomous inspection of large-scale infrastructure like bridges and pipelines, as well as providing real-time maps and guidance for search-and-rescue operations in disaster zones.

Annexure I – Weight & Battery Performance Analysis

Detailed Drone Weight Calculation (Dry Weight)

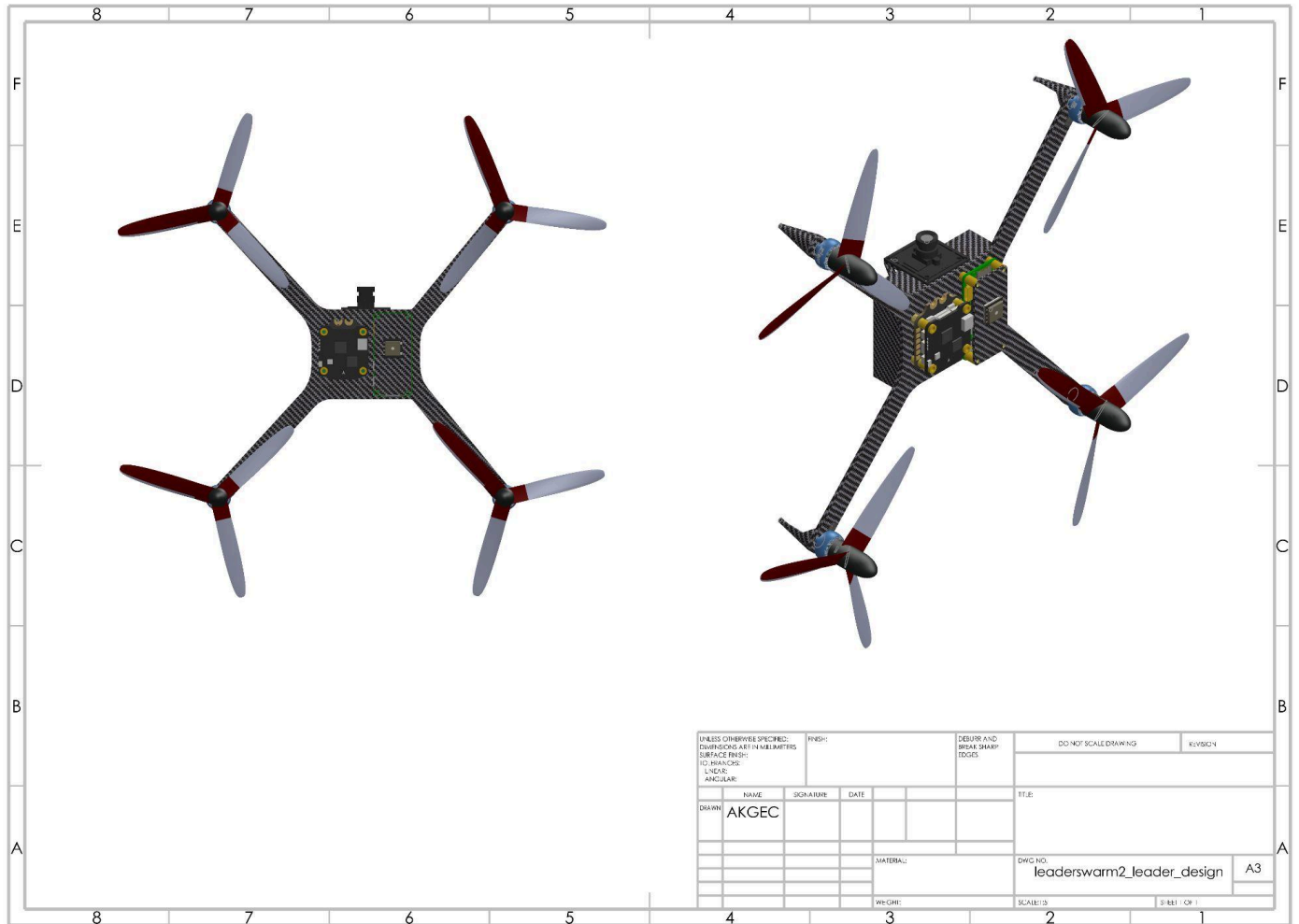
Subsystem	Components	Weight (g)
Propulsion System	Custom Carbon Fibre Frame , Motors (4×1404), Flight Controller (SpeedyBee)/ESC(35A 4-in-1 BLHeli_S ESC), Propellers	111 g
Smart Payload & Avionics	Raspberry Pi 4B, Camera Module, Benewake TFmini-S LiDAR, GPS Module, 5V BEC	70 g
Ancillary Components	Nylon Spacers, Hardware & Wiring	10 g
Total Dry Weight		191 g

The drone's total dry weight is 191 g.

Battery Performance Analysis

The 4S 18650 (Molicel P28A) battery pack is selected for its high capacity and ability to maintain an All-Up Weight (AUW) of under 500g while still ensuring stable flight.

- Battery Specification: The battery pack has a capacity of 2800 mAh and a nominal voltage of 14.8V (for a 4S configuration).
- All-Up Weight (AUW): With the addition of the battery pack, the total AUW of the drone is under 500g, ensuring it is compliant with competition constraints.
- Endurance Goal: The endurance goal of 15+ minutes is realistically achievable due to the high energy density of the battery pack. The drone’s thrust-to-weight ratio of approximately 4.8:1 ensures stable and responsive flight with the chosen components.



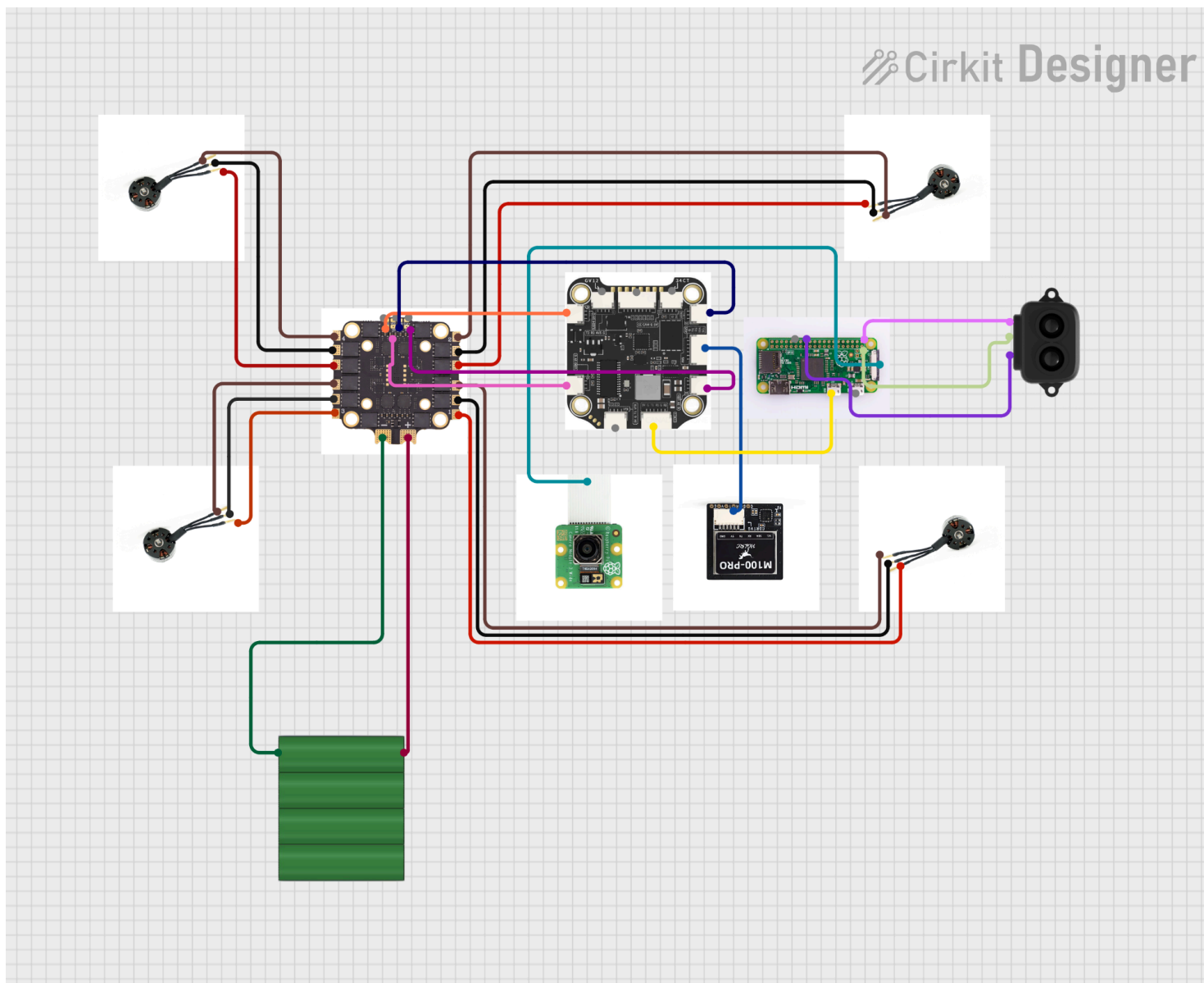
Please Follow the Drive link to view Proposed CAD Model in Browser

View: (https://drive.google.com/drive/folders/1QWU3l2qE9X-J_Pemt7thpzkdS879rJ8k?usp=sharing)

Annexure III – Cost Analysis

Component	Approximate Price (INR)
Raspberry Pi Camera Module 3	₹2,500
Speedbee F7 MINI Flight Controller(ESC INTEGRATED)	₹5874
HGLRC M100 Pro GPS	₹1900
Gemfan Flash 4052 Tri-Blade 3" Prop (4 Pack)	₹920
Battery Pack 4S (4× Molicel P28A 18650, 2800mAh)	₹1400
Benewake TFmini-S LiDAR	₹2,700
iFlight Xing2 1404 Motor (each)	₹6000
Raspberry Pi Zero 2 W	1500
Carbon Fiber Sheet (3mm thickness)	₹2,600 - ₹3,600 (300x300 to 300x200mm)

Annexure IV – Electronic Circuit Diagram



Annexure V – Additional Details

A. Expertise available with the Mentor:

Our mentor brings expertise in Embedded Systems, Wireless Sensor Networks, and Signal Processing, with experience in both research and applied domains. She has supervised projects involving real-time embedded solutions, wireless communication, and advanced signal processing techniques. Her guidance emphasizes practical understanding, innovation, and problem-solving, preparing mentees for academic excellence and industry readiness. With active involvement in research and consultancy projects, she provides real-world insights that bridge theory and practice. She is committed to nurturing talent and fostering future-ready skills in technology-driven fields.

B. List of Participation in past in any Robot Making Competitions:

EVENT NAME	VENUE	TYPE OF COMPETITION	TYPE OF ROBOT	ACHIEVEMENT
ROBOCON' 25	IIT DELHI	NATIONAL	SEMI AUTONOMOUS	AIR 12
AKTU ZONALS	KIET	UNIVERSITY	War Robot	3rd
I.R.O.C	URSC Bengaluru Campus	NATIONAL	Mars Drone Without GPS	Participants

C. Bank Details:

Name of the Bank	KOTAK MAHINDRA BANK LTD
Branch Name and Branch Code	Navyug Market
Address of the Bank with Pin code	30-31, NAVYUG MARKET, GHAZIABAD
Telephone no of the Bank	9899152827
Name of the Account Holder with Designation	Ajay Kumar Garg Engineering College
Account type	Saving
Account Number	508010250461
Bank Branch IFSC Code	KKBK0005295

D. Undertaking by the mentor



AJAY KUMAR GARG ENGINEERING COLLEGE

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Mentor Undertaking Format

I, Dr. Himani Garg Mentor on behalf of my team is authorized to give undertaking that on selection of our team at Level 1 (Ideation Stage), we assure and commit to participate in the subsequent levels and to submit Level 2 (Proof of Concept) and Level 3 (Proto type) robot as per guidelines of GUJCOST without fail, otherwise GUJCOST will take necessary action to recover the fund if disbursed for any stage.

Himani 25/8/25
Dr. Himani Garg
Professor (ECE)

A. H.
25/8/25

Dr. Hemant Ahuja
Director

Director
Ajay Kumar Garg Engg. College
Ghaziabad

Recipient of the Academic Excellence Award for Best Engineering College in UPTU
from H.E. The Governor of U.P. for two consecutive years